RTOS Benchmark on STM32H750B-DK

Phase 1 - DWT-only Publishable Campaign

Four-test latency comparison of ChibiOS RT 7.0.6, FreeRTOS V11.3.0 and Zephyr 4.4.0 running at 480 MHz (VOS0, FLASH WS=4, I-Cache + D-Cache enabled) on the STM32H750B-DK Discovery Kit.

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Abstract

This report documents the Phase 1 DWT-only publishable benchmark campaign of three open-source real-time operating systems on a single, fully neutral STM32H750B-DK target. The three kernels are compared under identical hardware conditions: CPU clock 480 MHz, voltage scaling VOS0, FLASH_ACR = 0x34, I-Cache + D-Cache enabled, identical GCC 14.2 toolchain and -O2 -fomit-frame-pointer -falign-functions=16 effective flags. Two publishable profiles are reported. The headline profile is **realistic_tickless** (tickless idle ON, WFI ON), a representative low-power, shipping-like configuration; **fair_perf** (periodic tick, no WFI) is published as a controlled-isolation baseline that exposes scheduler and primitive cost without idle re-arm effects. Latencies for the four tests T1-T4 are measured inside the firmware via the Cortex-M7 DWT cycle counter. Each (RTOS, profile) combination has been validated against five independent firmware loads, with publication-gate verification of clock, cache, flash, memory placement and priority-inheritance correctness; the run_spread metric is zero cycles on every (RTOS, test) cell, demonstrating fully reproducible results across reloads. Under realistic_tickless, ChibiOS showed the lowest median latency in all four tests (T1 median 306 cycles vs FreeRTOS 846 and Zephyr 581); the fair_perf baseline preserves the same ranking. Between the two profiles the FreeRTOS T1 median rises from 440 to 846 cycles (+406) — a measured cost of the FreeRTOS tickless wake-up re-arm path under this published configuration; ChibiOS (+9) and Zephyr (-43) move much less. Mutex priority-inheritance is correct in all three kernels with 500/500 events verified per RTOS per profile. FreeRTOS and Zephyr are excellent RTOS projects with different design goals; this comparison focuses only on real-time latency under the tested conditions.

Test environment

Hardware

Property	Value
Board	STM32H750B-DK Discovery Kit
MCU	STM32H750XBH6, silicon revision V
Core	Cortex-M7 with double-precision FPU
External oscillator (HSE)	25 MHz crystal
CPU clock	480 MHz (PLL1 from HSE)
Voltage scaling	VOS0 (required for 480 MHz)
FLASH_ACR	0x00000034 (LATENCY = 4 WS, programming delay 2)
I-Cache / D-Cache	ON (production-realistic)
Internal Flash	128 KB (no bootloader, application fits)
RAM	1 MB (64K ITCM + 128K DTCM + 864K AXI/AHB SRAM)
Debug / VCP	ST-LINK V3E onboard, USART3 PB10/PB11, 115200 8N1

Toolchain and compile flags

Tool	Version / Setting
GCC	arm-none-eabi-gcc 14.2.Rel1 (Arm GNU Toolchain)
Make	GNU Make 4.3 (MSYS2)
OpenOCD	0.12.0+dev (xPack)
Optimization	-O2 (no -Os, no -O3, no LTO)
ADR-009 effective flags	-fomit-frame-pointer -falign-functions=16
C standard	C11 application code, C99 portable modules
Cross-RTOS verification	compile_commands.json read-back audit (per port)

RTOS versions

RTOS	Version	Source
ChibiOS RT	7.0.6	ChibiOS 21.11.5, git tag ver21.11.5 (upstream, no fork)
FreeRTOS Kernel	V11.3.0	tag V11.3.0 (latest stable, March 2026)
Zephyr	4.4.0	tag v4.4.0 (latest stable, April 2026)
STM32CubeH7 HAL	v1.12.1	for FreeRTOS variant only

All three kernels are pinned to their current stable release: ChibiOS 21.11.5 (RT 7.0.6, git tag ver21.11.5), FreeRTOS V11.3.0 and Zephyr 4.4.0. ChibiOS 21.11.x is the maintained stable line and 21.11.5 is its latest point release, so the comparison uses the current stable version of each kernel rather than a frozen older tag. The published firmware was captured at submodule commit 78a2ddd2, which is ver21.11.5 plus the removal of one non-compiled SBOM file; the source pin was subsequently aligned to the named tag ver21.11.5, and a clean rebuild from the pinned tree reproduces every published ELF and MAP SHA-256 byte-for-byte.

Clock-tree witness (banner of run 01)

The firmware banner of every run prints back the values of RCC, PWR and FLASH registers immediately after boot, so that the publishable claims of 480 MHz / VOS0 / WS=4 are anchored to actually measured silicon state, not just configuration sources. The excerpt below is taken from *chibios_fair_perf_run01.banner.txt* and is representative.

Register / field	Value
SystemClock	`480000000 Hz`
VOS level	`VOS0`
VOSRDY	`READY`
FLASH_ACR	`0x00000034`
RCC_PLLCKSELR	
RCC_PLLCFGR	`0x01ff0ff9`
RCC_PLL1DIVR	`0x010e02bf`
RCC_D1CFGR	`0x00000048`
PWR_D3CR	`0x0000e000`
SCB->CCR	`0x00070200`
ICache	`ON`
DCache	`ON`
Tick rate	`1000 Hz`
Optimization	`-O2  LTO=no`

Methodology

The four tests

ID	Name	What is measured
T1	IRQ -> thread latency	TIM2 compare match triggers a hardware-routed IRQ; a high-priority worker thread is unblocked. DWT counter is sampled at ISR entry (A1) and again as soon as the worker thread resumes execution (A4). The published metric is the cycle delta A4 - A1, i.e. the path ISR_ENTRY -> THREAD_RUNNING.
T2	Thread handoff	A tester thread suspends/resumes a higher-priority target via the native suspend/resume primitive (the target blocks, the tester wakes it). Pure thread-to-thread context-switch cost, with the scheduler hot in cache.
T3	Mutex lock / unlock, uncontended	A single thread locks and immediately unlocks a mutex in a tight loop. No contention. Measures the cost of the fast path through the mutex primitive.
T4	Mutex contended + priority inheritance	Three threads (LOW, MEDIUM, HIGH). LOW owns the mutex, HIGH waits on it, MEDIUM tries to disturb. The scheduler must boost LOW to HIGH's priority and keep MEDIUM excluded for the duration of the critical section. The protocol verifies the PI invariant with 100 one-shot scenarios per run.

Cross-RTOS equivalence. The benchmark defines the same external scenario for all three kernels and implements it with each RTOS's most direct native primitive. The source code is therefore not identical across ChibiOS, FreeRTOS and Zephyr; the comparison is scenario-equivalent under identical hardware, clock, compiler and measurement conditions.

Primitive used per test

The exact native call exercised inside the measured DWT window for each (test, RTOS). For T1 and T2 these are the wake / handoff primitives themselves; the readiness handshakes used only to start each test are excluded.

Test	ChibiOS	FreeRTOS	Zephyr
T1	chThdSuspendS / chThdResumeI	ulTaskNotifyTake / vTaskNotifyGiveFromISR + portYIELD_FROM_ISR	k_sem_take / k_sem_give
T2	chSchGoSleepS / chSchWakeupS	vTaskSuspend / vTaskResume	k_thread_suspend / k_thread_resume
T3	chMtxLock / chMtxUnlock	xSemaphoreCreateMutexStatic; xSemaphoreTake / Give	k_mutex_lock / k_mutex_unlock
T4	chMtxLock / chMtxUnlock (PI)	xSemaphoreCreateMutexStatic (PI); xSemaphoreTake / Give	k_mutex_lock / k_mutex_unlock (PI)

Measurement protocol

All latencies are measured using the Cortex-M7 DWT cycle counter (32-bit free-running, no external timing). DWT overhead is 4 cycles per pair of samples (0.008 us at 480 MHz) and is documented in the firmware banner. T1, T2 and T3 run for 1000 warmup iterations (discarded) followed by 10 000 measured iterations. T4 runs 100 one-shot PI scenarios without warmup (ADR-013 explicit exception, because PI must be observed on the very first event of a fresh kernel state, not after a steady-state warmup). Samples are stored in pre-allocated static arrays placed in AXI-SRAM (memory placement verified per run via the *bench_addr* banner table). No dynamic allocation is used by the benchmark code in any RTOS.

Campaign protocol (ADR-013)

For every (RTOS, profile) combination the firmware is fully rebuilt, flashed and run five independent times (run01 through run05). The hash of the ELF and MAP produced by each rebuild is recorded and verified to be identical across runs of the same target (homogeneity check). The serial collector discards any stale bytes from the ST-Link VCP receive buffer before listening, so that no row from a previous firmware can leak into the current capture. The publication gate rejects a run unless ALL of the following hold: TIM2 setup readback matches the expected state; memory placement banner confirms every benchmark object lives in AXI-SRAM; iteration sequences are dense and ordered; T1/T2/T3 produce exactly 10 000 valid rows; T4 produces exactly 100 PI scenarios with 100/100 passes; ELF and MAP SHA-256 match the campaign lock. The reported median per (RTOS, test) is the median-of-medians across the five runs; jitter is max - min in cycles; run_spread is the cycle span of the per-run medians (a strictly internal stability metric).

On the interpretation of the *max (cyc)* column in the fair_perf profile: a fair_perf run does not suppress asynchronous interrupts (notably the 1 kHz SysTick at 480 MHz), so a single iteration's measured cycle window can capture both the native operation and any asynchronous ISR that happens to land inside the same DWT sample. The primary cross-RTOS hot-path comparison is therefore based on **median, p95 and p99**, while *max* is reported for transparency rather than presented as the pure native operation cost. realistic_tickless suppresses the periodic tick when the system is idle but a CPU-bound loop in any RTOS can still be interrupted by other asynchronous events.

Scope and non-claims (Phase 1)

Phase 1 publishes the DWT cycle-delta A4 - A1 as the headline figure for T1. This metric covers the path from ISR entry to the worker thread regaining execution, **excluding** the silicon hardware-event-to-ISR-entry portion (timer compare match, NVIC arbitration, exception stacking, vector fetch and ISR prologue). The figures in this report must not be quoted as the absolute external IRQ-to-thread latency that an external logic analyzer would observe at GPIO pins; see ADR-015 for the Phase 2 dual-source plan where the analyzer-based metric is introduced alongside the DWT one.

Benchmark conditions

The comparison below is valid only under the exact conditions stated here.

Item	Value
Board / MCU	STM32H750B-DK / STM32H750XBH6 (Cortex-M7, DP-FPU)
CPU clock / cache / flash	480 MHz (VOS0); I-Cache + D-Cache ON; FLASH_ACR = 0x34 (4 wait states)
Compiler / optimization	arm-none-eabi-gcc 14.2.Rel1; -O2 -fomit-frame-pointer -falign-functions=16; no LTO
RTOS versions / source	ChibiOS 21.11.5 / RT 7.0.6 (git tag ver21.11.5); FreeRTOS V11.3.0 (tag V11.3.0); Zephyr 4.4.0 (tag v4.4.0)
Profile configuration	realistic_tickless (headline): tickless ON, WFI ON. fair_perf (controlled-isolation baseline): tickless OFF, WFI OFF. No asserts, debug or logging in the measured path.
Measurement method	Cortex-M7 DWT CYCCNT cycle deltas inside the firmware. Phase 1 publishes A4 - A1 (ISR entry -> thread running); the external hardware-event-to-ISR portion is Phase 2 (logic analyzer).
Iterations	1000 warmup (discarded) + 10000 valid per run for T1/T2/T3; 100 one-shot scenarios for T4; 5 validated firmware loads per (RTOS, profile).
Repository	https://github.com/chibilogic/rtos-benchmark/tree/phase1-v1.0
Raw logs	https://raw.githubusercontent.com/chibilogic/rtos-benchmark/phase1-v1.0/published-logs/phase1/phase1-raw-logs-467348c9306d.zip
Scope	Results apply only to this benchmark configuration.

Results - profile realistic_tickless

Profile semantics: realistic_tickless (tickless ON, WFI ON) — the headline representative profile (low-power, shipping-like: tickless idle + WFI enabled). All other publication invariants (480 MHz, VOS0, FLASH_ACR=0x34, I-Cache + D-Cache ON, -O2 effective flags, DWT measurement) are unchanged.

Aggregate results

Median, percentile, jitter and run_spread for each (RTOS, test) cell, computed across 10 000 iterations per run x 5 runs (T1/T2/T3) or 100 scenarios per run x 5 runs (T4). run_spread = 0 cycles on every cell means the median of each individual run was bit-identical to the median of every other run for that (RTOS, test) -- a per-run stability metric, NOT an absence of variance: the p95 / p99 / max columns below capture the real distribution (e.g. an asynchronous SysTick tick landing in-sample appears as a max well above the median), so the flat run_spread reflects reproducible medians, not over-clean data.

RTOS	Test	median (cyc)	median (us)	p95 (cyc)	p99 (cyc)	p99 (us)	max (cyc)	jitter	run_spread
ChibiOS	t1_irq	306	0.637	322	324	0.675	328	42	0
ChibiOS	t2_handoff	95	0.198	96	96	0.200	110	16	0
ChibiOS	t3_mtx_uncont	62	0.129	64	64	0.133	77	15	0
ChibiOS	t4_mtx_pi	200	0.417	200	219	0.456	219	19	0
FreeRTOS	t1_irq	846	1.762	848	850	1.771	1027	202	0
FreeRTOS	t2_handoff	376	0.783	397	417	0.869	566	208	0
FreeRTOS	t3_mtx_uncont	195	0.406	195	196	0.408	387	194	0
FreeRTOS	t4_mtx_pi	612	1.275	616	641	1.335	641	29	0
Zephyr	t1_irq	581	1.210	582	614	1.279	616	41	0
Zephyr	t2_handoff	424	0.883	424	424	0.883	439	17	0
Zephyr	t3_mtx_uncont	125	0.260	125	125	0.260	140	15	0
Zephyr	t4_mtx_pi	677	1.410	686	715	1.490	715	38	0

Cross-RTOS comparison

Test	ChibiOS (cyc)	FreeRTOS (cyc)	Zephyr (cyc)	ChibiOS (us)	FreeRTOS (us)	Zephyr (us)	Lowest median (this test)
t1_irq	**306**	846	581	0.637	1.762	1.210	**ChibiOS**
t2_handoff	**95**	376	424	0.198	0.783	0.883	**ChibiOS**
t3_mtx_uncont	**62**	195	125	0.129	0.406	0.260	**ChibiOS**
t4_mtx_pi	**200**	612	677	0.417	1.275	1.410	**ChibiOS**

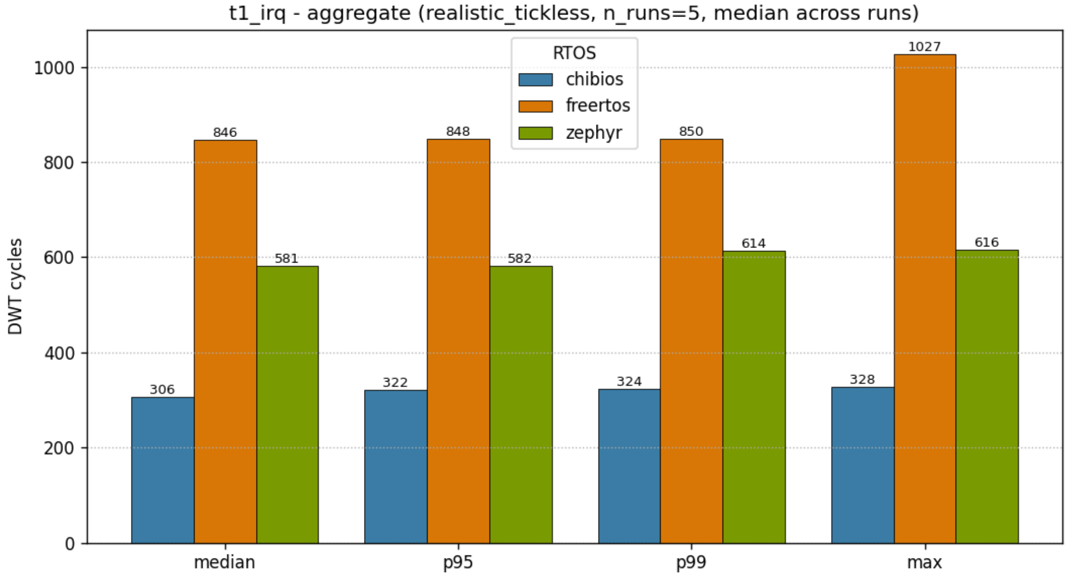
Method: median-of-medians over five validated firmware loads; DWT CYCCNT cycle deltas; lower is better for these latency tests. Results apply only to the benchmark configuration stated in this report.

Priority-inheritance proof (T4)

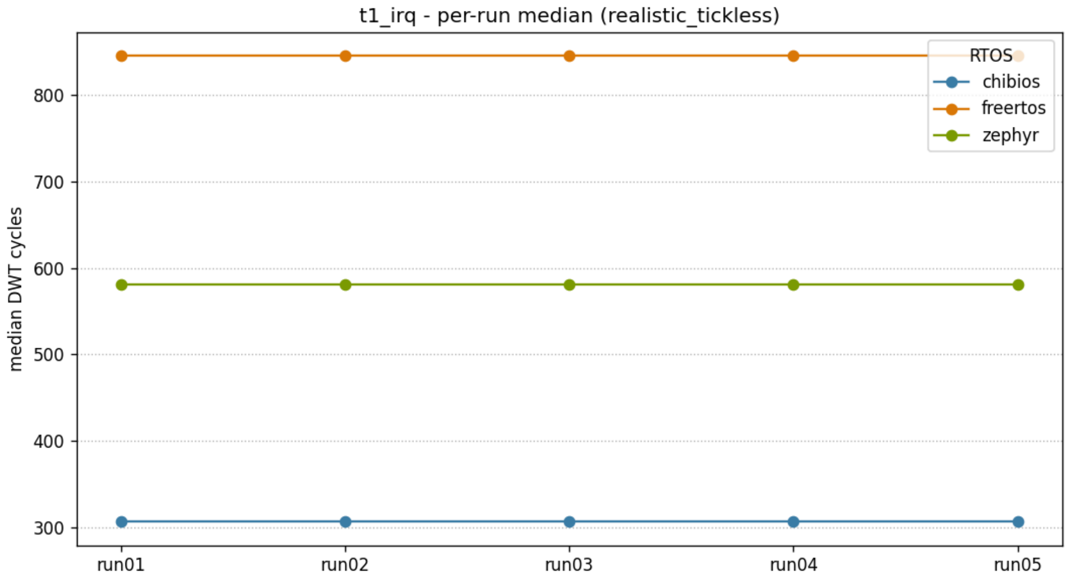
RTOS	PI scenarios	PI passed	Result
ChibiOS	500	500	**PASS**
FreeRTOS	500	500	**PASS**
Zephyr	500	500	**PASS**

Per-test plots

T1 - IRQ -> thread latency

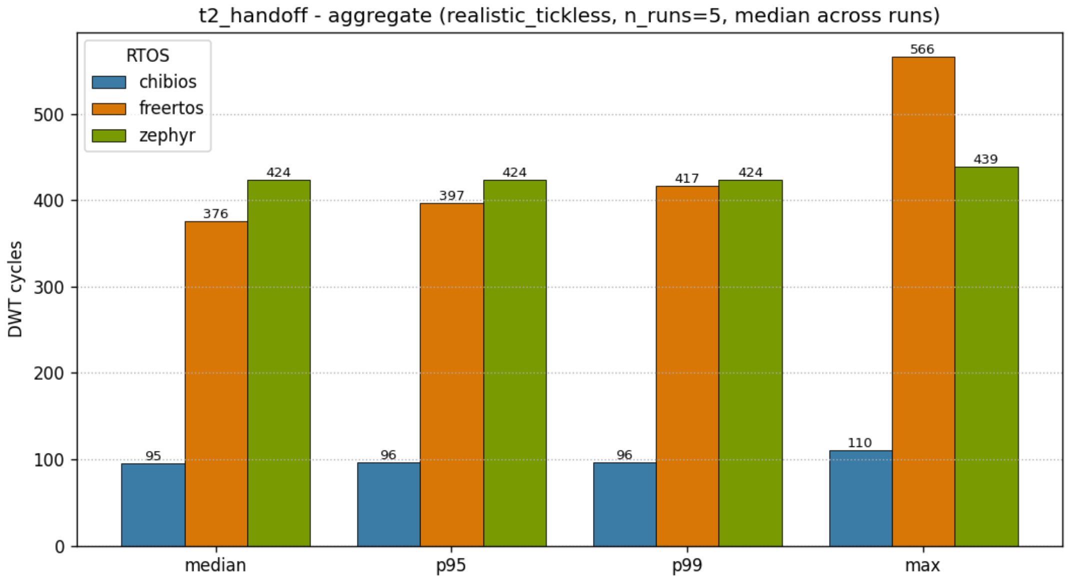


Aggregate distribution - realistic_tickless / t1_irq

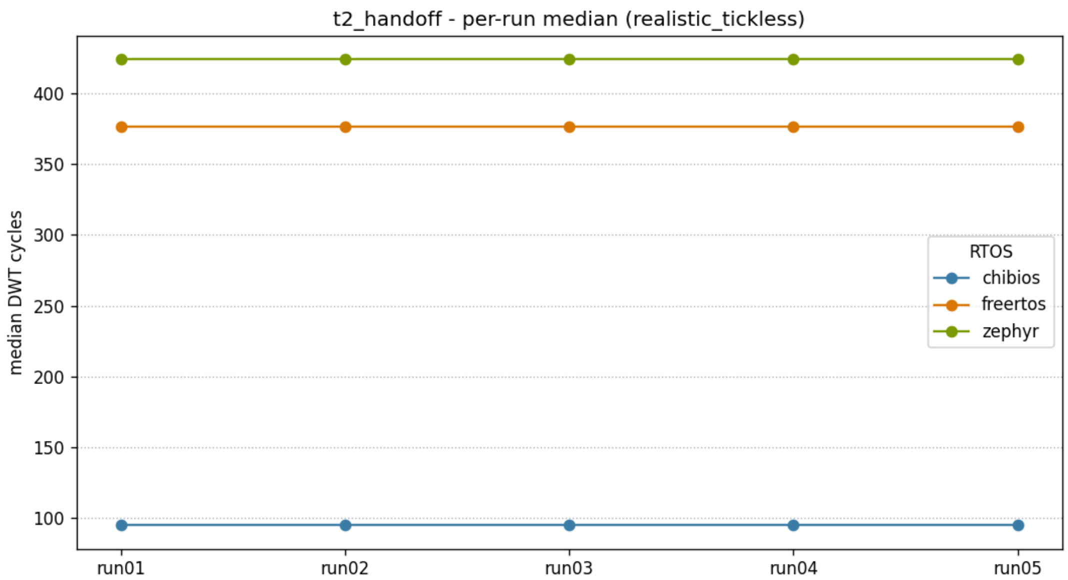


Per-run medians - realistic_tickless / t1_irq

T2 - Thread handoff

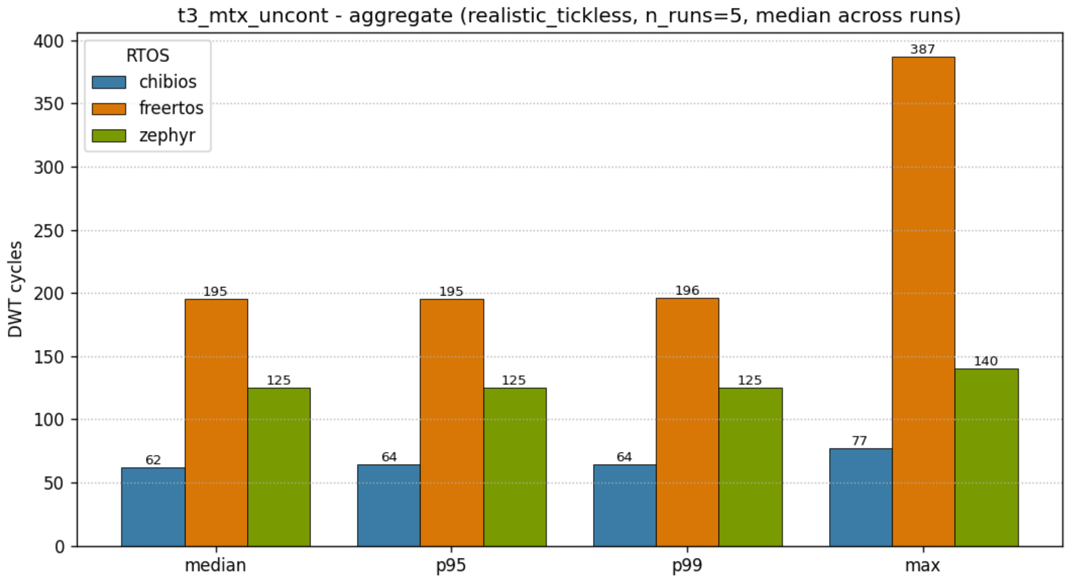


Aggregate distribution - realistic_tickless / t2_handoff

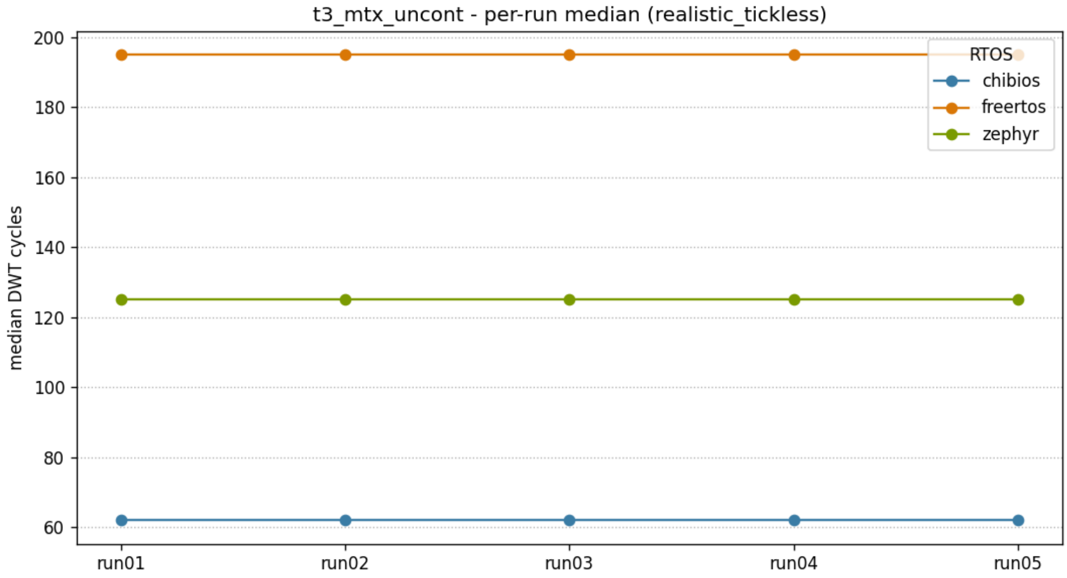


Per-run medians - realistic_tickless / t2_handoff

T3 - Mutex uncontended

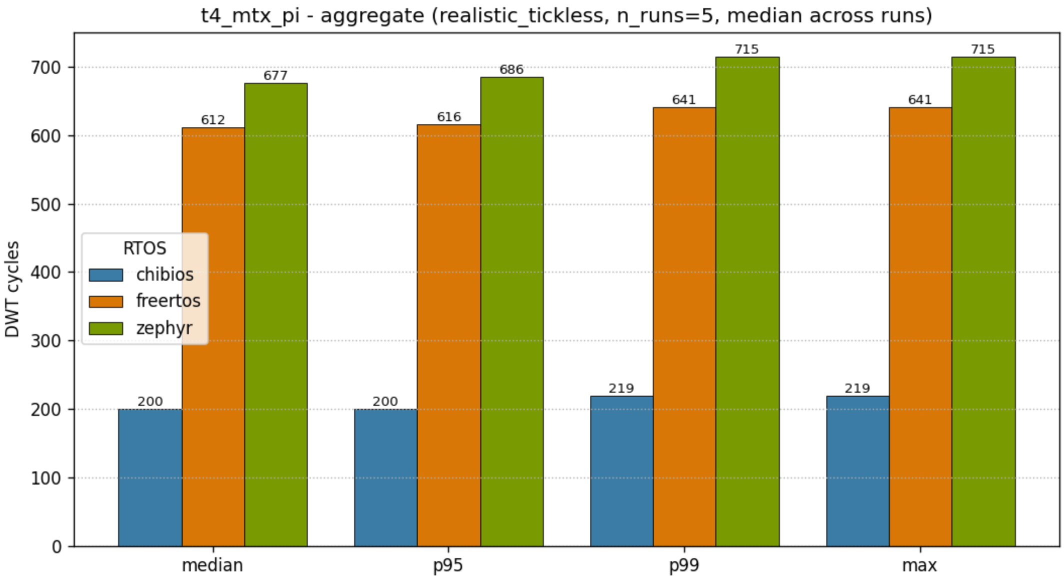


Aggregate distribution - realistic_tickless / t3_mtx_uncont

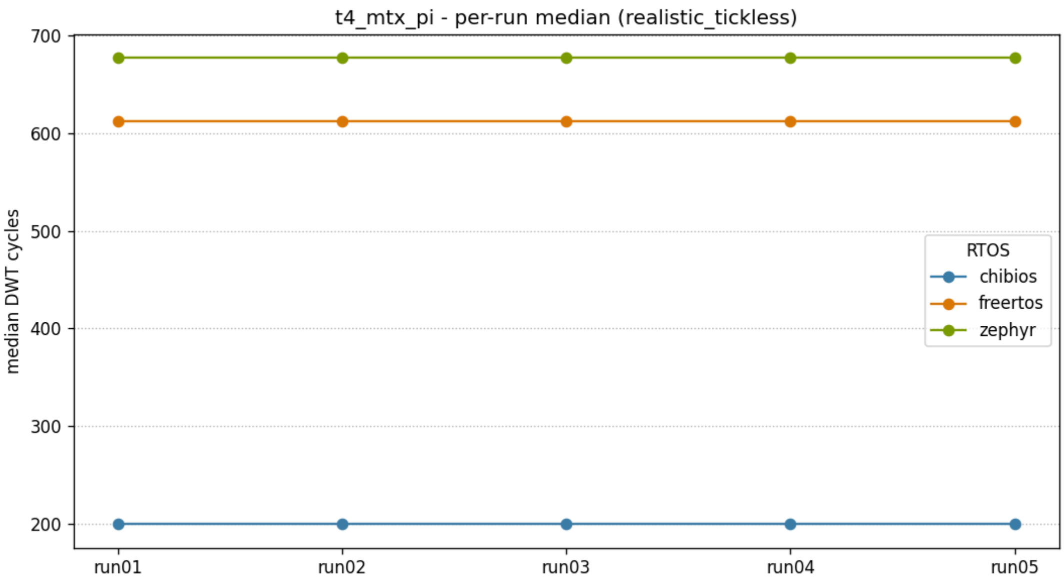


Per-run medians - realistic_tickless / t3_mtx_uncont

T4 - Mutex contended + priority inheritance

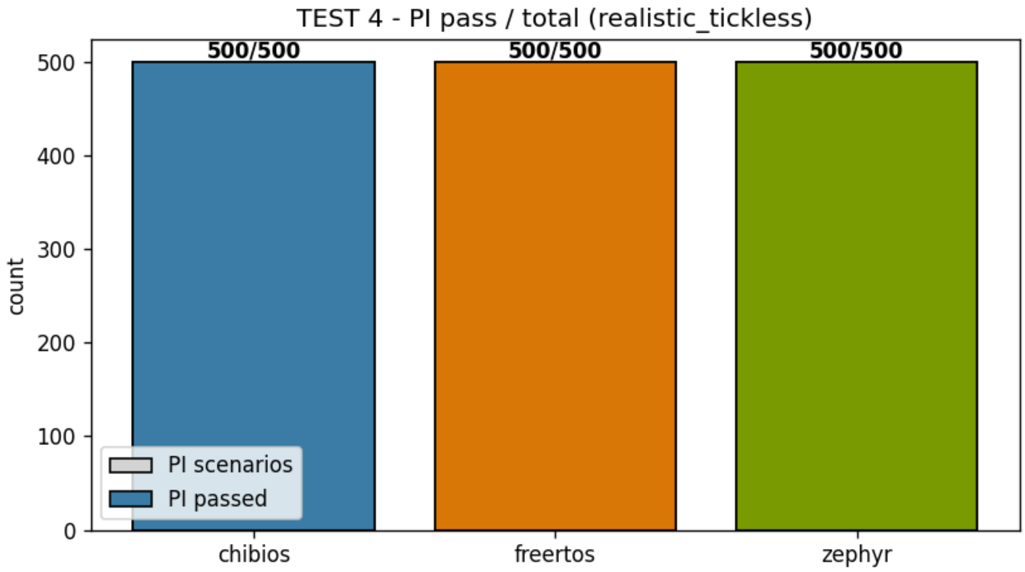


Aggregate distribution - realistic_tickless / t4_mtx_pi



Per-run medians - realistic_tickless / t4_mtx_pi

Priority-inheritance event distribution



T4 priority-inheritance scenarios - realistic_tickless

Results - profile fair_perf

Profile semantics: fair_perf (tickless OFF, WFI OFF) — the controlled-isolation baseline (periodic tick, no WFI) that exposes scheduler / primitive cost without idle re-arm effects. All other publication invariants (480 MHz, VOS0, FLASH_ACR=0x34, I-Cache + D-Cache ON, -O2 effective flags, DWT measurement) are unchanged.

Aggregate results

Median, percentile, jitter and run_spread for each (RTOS, test) cell, computed across 10 000 iterations per run x 5 runs (T1/T2/T3) or 100 scenarios per run x 5 runs (T4). run_spread = 0 cycles on every cell means the median of each individual run was bit-identical to the median of every other run for that (RTOS, test) -- a per-run stability metric, NOT an absence of variance: the p95 / p99 / max columns below capture the real distribution (e.g. an asynchronous SysTick tick landing in-sample appears as a max well above the median), so the flat run_spread reflects reproducible medians, not over-clean data.

RTOS	Test	median (cyc)	median (us)	p95 (cyc)	p99 (cyc)	p99 (us)	max (cyc)	jitter	run_spread
ChibiOS	t1_irq	297	0.619	303	303	0.631	318	34	0
ChibiOS	t2_handoff	93	0.194	94	94	0.196	110	18	0
ChibiOS	t3_mtx_uncont	62	0.129	64	64	0.133	403	341	0
ChibiOS	t4_mtx_pi	200	0.417	200	220	0.458	220	20	0
FreeRTOS	t1_irq	440	0.917	447	447	0.931	459	27	0
FreeRTOS	t2_handoff	370	0.771	379	385	0.802	551	197	0
FreeRTOS	t3_mtx_uncont	202	0.421	216	223	0.465	429	233	0
FreeRTOS	t4_mtx_pi	599	1.248	599	632	1.317	632	33	0
Zephyr	t1_irq	624	1.300	630	635	1.323	649	54	0
Zephyr	t2_handoff	425	0.885	425	425	0.885	641	221	0
Zephyr	t3_mtx_uncont	124	0.258	124	124	0.258	442	318	0
Zephyr	t4_mtx_pi	722	1.504	722	766	1.596	766	44	0

Cross-RTOS comparison

Test	ChibiOS (cyc)	FreeRTOS (cyc)	Zephyr (cyc)	ChibiOS (us)	FreeRTOS (us)	Zephyr (us)	Lowest median (this test)
t1_irq	**297**	440	624	0.619	0.917	1.300	**ChibiOS**
t2_handoff	**93**	370	425	0.194	0.771	0.885	**ChibiOS**
t3_mtx_uncont	**62**	202	124	0.129	0.421	0.258	**ChibiOS**
t4_mtx_pi	**200**	599	722	0.417	1.248	1.504	**ChibiOS**

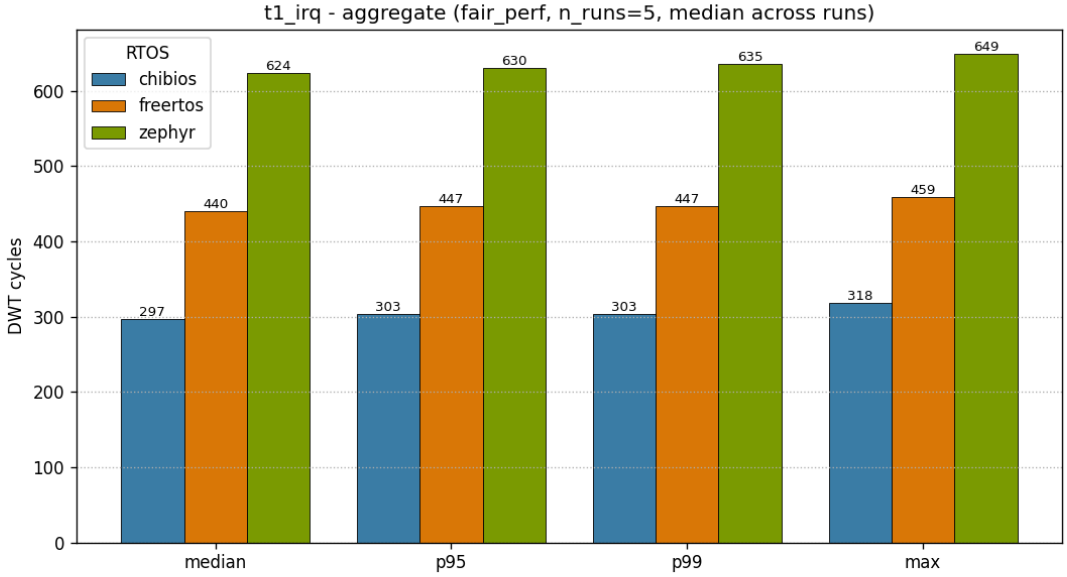
Method: median-of-medians over five validated firmware loads; DWT CYCCNT cycle deltas; lower is better for these latency tests. Results apply only to the benchmark configuration stated in this report.

Priority-inheritance proof (T4)

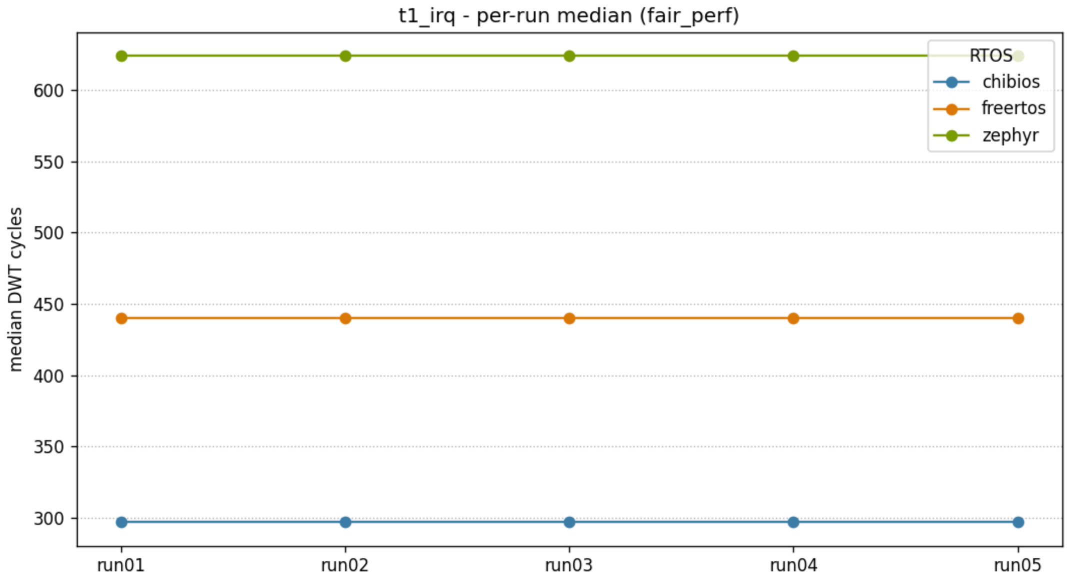
RTOS	PI scenarios	PI passed	Result
ChibiOS	500	500	**PASS**
FreeRTOS	500	500	**PASS**
Zephyr	500	500	**PASS**

Per-test plots

T1 - IRQ -> thread latency

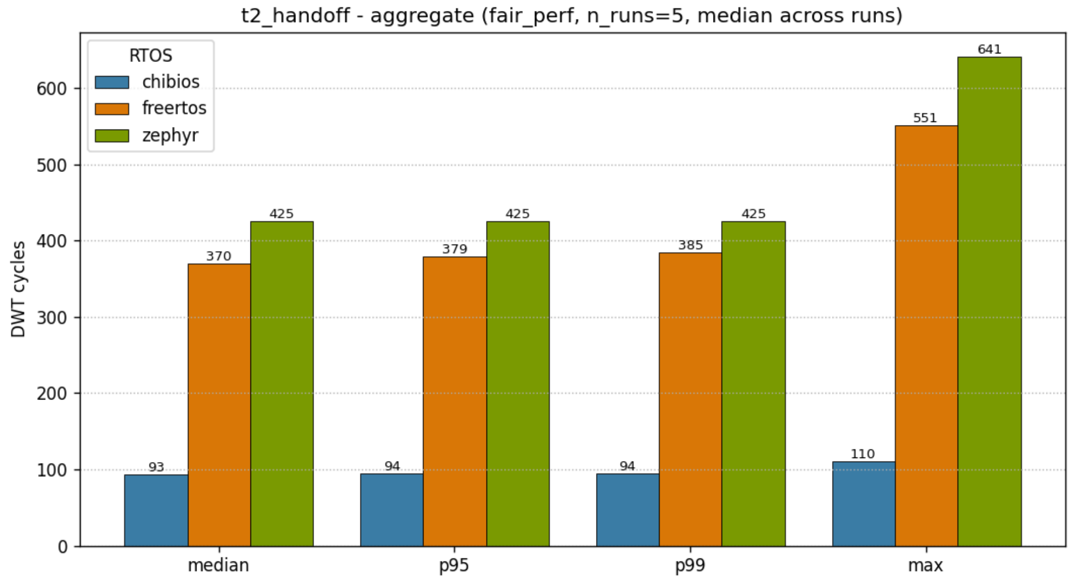


Aggregate distribution - fair_perf / t1_irq

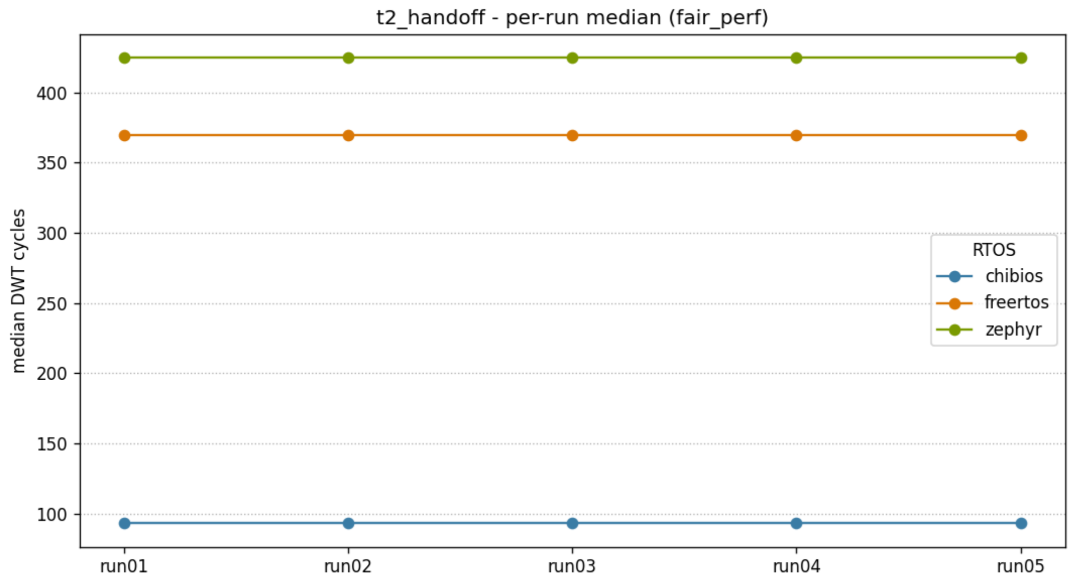


Per-run medians - fair_perf / t1_irq

T2 - Thread handoff

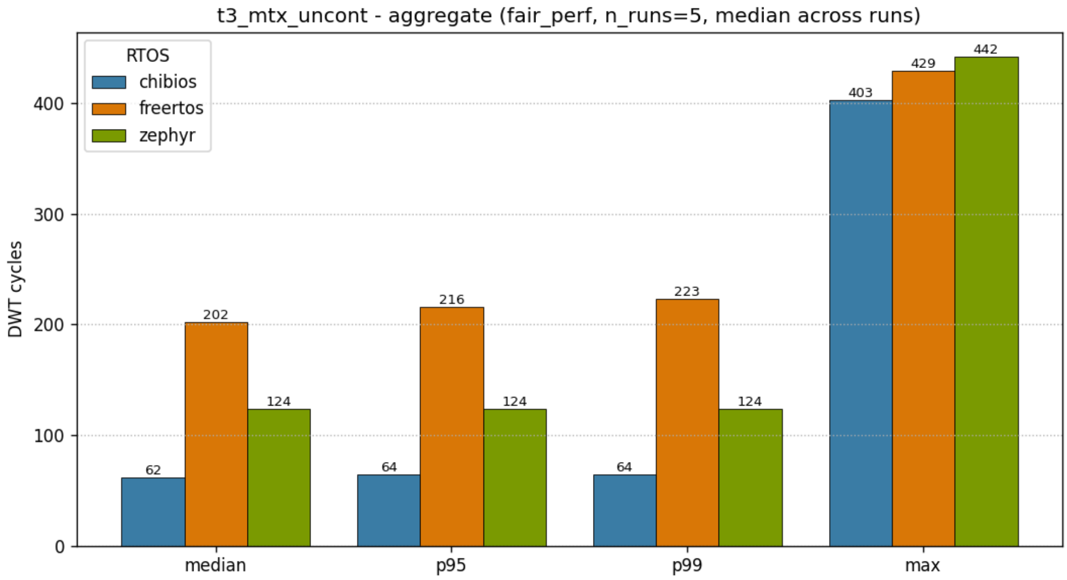


Aggregate distribution - fair_perf / t2_handoff

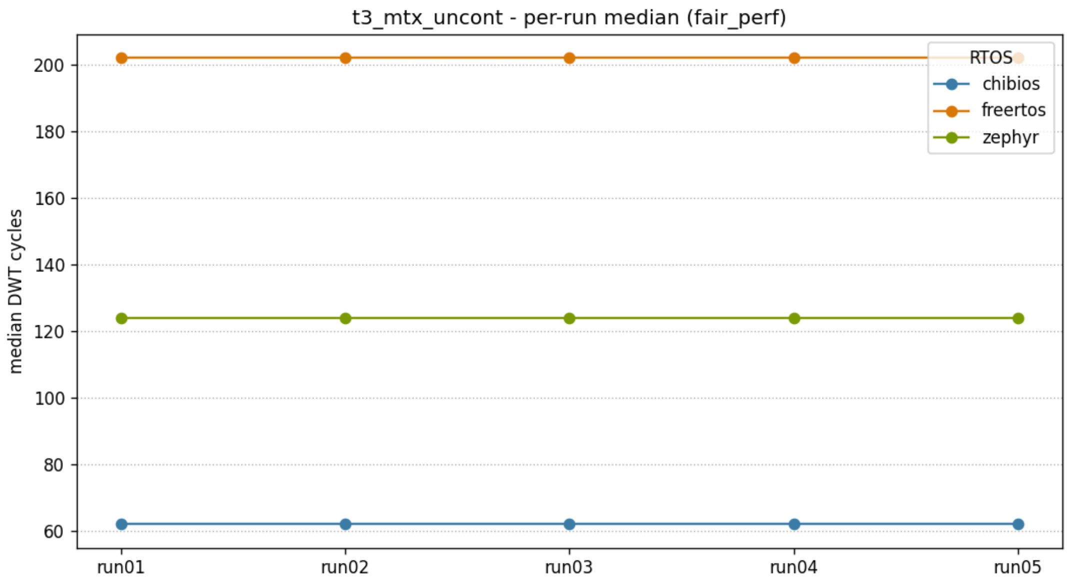


Per-run medians - fair_perf / t2_handoff

T3 - Mutex uncontended

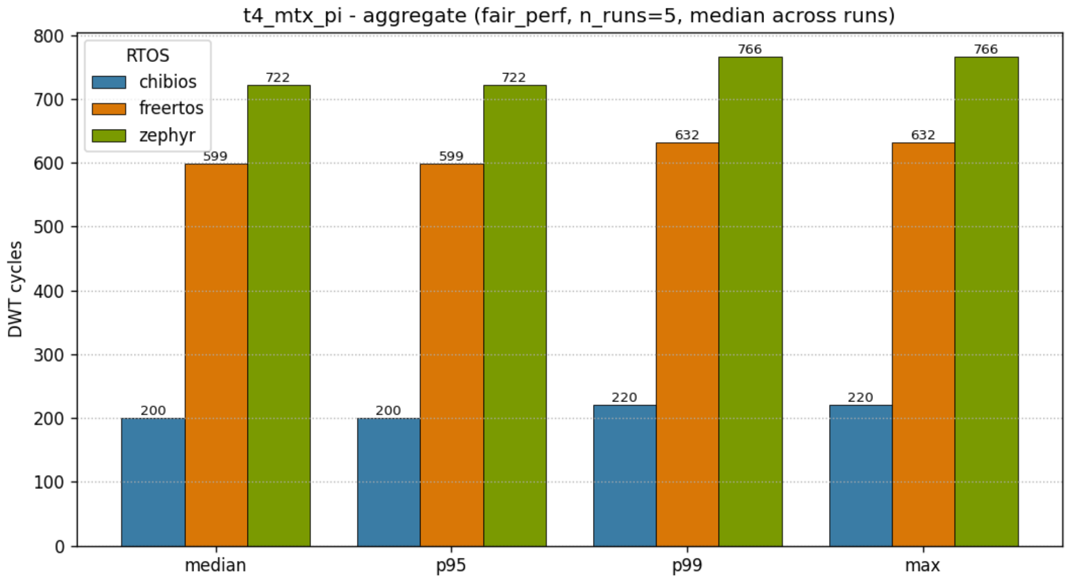


Aggregate distribution - fair_perf / t3_mtx_uncont

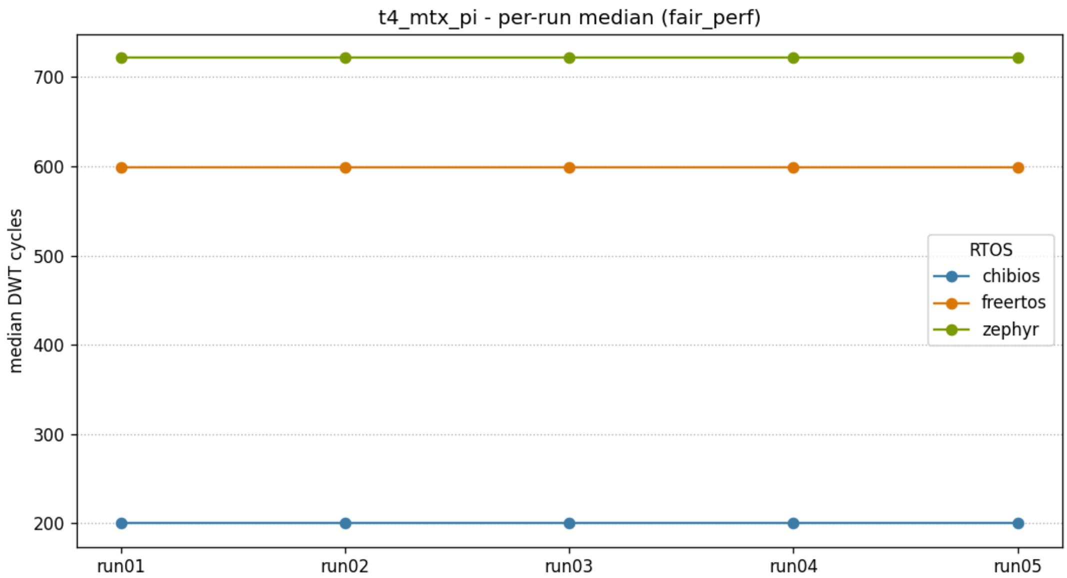


Per-run medians - fair_perf / t3_mtx_uncont

T4 - Mutex contended + priority inheritance

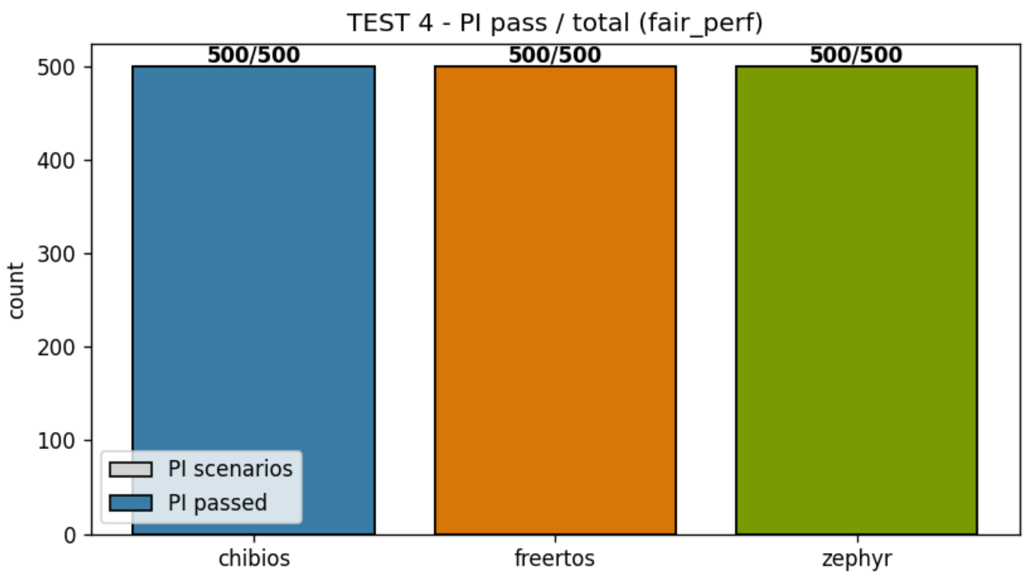


Aggregate distribution - fair_perf / t4_mtx_pi



Per-run medians - fair_perf / t4_mtx_pi

Priority-inheritance event distribution



T4 priority-inheritance scenarios - fair_perf

Cross-profile delta (baseline fair_perf -> headline realistic_tickless)

Per-cell difference of the median latency from the controlled-isolation baseline (fair_perf) to the headline profile (realistic_tickless). The arithmetic direction is baseline-to-headline, so a positive number means the headline profile is slower. The notable swing is T1 for FreeRTOS, where enabling the tickless idle path adds a wake-up re-arm overhead specific to the FreeRTOS path under this published configuration.

RTOS	Test	fair_perf median (cyc)	realistic_tickless median (cyc)	delta (cyc)	delta (us)
ChibiOS	t1_irq	297	306	+9	+0.019
ChibiOS	t2_handoff	93	95	+2	+0.004
ChibiOS	t3_mtx_uncont	62	62	+0	+0.000
ChibiOS	t4_mtx_pi	200	200	+0	+0.000
FreeRTOS	t1_irq	440	846	**+406**	**+0.846**
FreeRTOS	t2_handoff	370	376	+6	+0.012
FreeRTOS	t3_mtx_uncont	202	195	-7	-0.015
FreeRTOS	t4_mtx_pi	599	612	+13	+0.027
Zephyr	t1_irq	624	581	-43	-0.090
Zephyr	t2_handoff	425	424	-1	-0.002
Zephyr	t3_mtx_uncont	124	125	+1	+0.002
Zephyr	t4_mtx_pi	722	677	-45	-0.094

Firmware footprint

Static firmware size indicators extracted from the manifest-bound publication ELF of each (RTOS, profile) - the same binary whose SHA-256 is locked in the campaign manifest and listed in the Integrity section. Metric definitions are binding per ADR-023.

Code size = .text + .rodata (read-only code and constants in flash). **Static RAM used** = .data + .bss + noinit, with the ChibiOS .heap linker reservation excluded (CH_CFG_USE_HEAP=FALSE, no allocation) and the per-RTOS committed main/process stacks added where they are not already a labelled section. **Tail reservation** is informational only: spare RAM held for the descending main stack (FreeRTOS), runtime k_thread_create stacks (Zephyr) or the linker heap reservation (ChibiOS); it is NOT part of the static-RAM figure.

fair_perf (tickless OFF, WFI OFF)

RTOS	code size (B)	static RAM used (B)	tail reservation (B, info)
ChibiOS	**24612**	**229984**	298400
FreeRTOS	31716	252776	276632
Zephyr	45508	236185	288000

Computed from results/raw/<rtos>_fair_perf_run01.elf; every ELF SHA-256 matches the campaign lock.

realistic_tickless (tickless ON, WFI ON)

RTOS	code size (B)	static RAM used (B)	tail reservation (B, info)
ChibiOS	**25256**	**229984**	298400
FreeRTOS	32396	252784	276624
Zephyr	45936	236193	288000

Computed from results/raw/<rtos>_realistic_tickless_run01.elf; every ELF SHA-256 matches the campaign lock.

libc disclaimer (ADR-009): the three RTOSes intentionally link three different C libraries - ChibiOS newlib (full), FreeRTOS newlib-nano, Zephyr picolibc. The code-size figure therefore bundles kernel + HAL + libc and is not a pure kernel-text comparison; a per-archive breakdown is a planned follow-up.

Integrity and reproducibility

The publication gate is enforced at three layers: (1) per-run validation, captured in the **.validated.json* manifest written by the collector alongside the CSV; (2) per-campaign lock, written by the lab campaign script after every (RTOS, profile) is complete; (3) build reproducibility, expressed as ELF / MAP SHA-256 homogeneity across the five runs of a given target. All numbers in this report come exclusively from validated runs.

Campaign locks - ELF and MAP SHA-256

fair_perf

RTOS	Artefact	SHA-256 (truncated)
ChibiOS	ELF	`18ad62d06bbd856f2b662c97d13a98db53f46715c0063ebb...`
	MAP	`0ec2c1b78ac7284e07e3f1ce855236ad1d52a248bd98b81c...`
FreeRTOS	ELF	`8410f98d6dc145805975eb65208f00fdf9002c82b4af5a77...`
	MAP	`d0f8a290dc5d88c2212e0e37e4e5b8fcd469fac32e877ecf...`
Zephyr	ELF	`0e175ae3127c194b2f7528622e82574aa39d7951c5654261...`
	MAP	`988ea68c24b02860470f907d6bb4035ea158fb363b09a81e...`

realistic_tickless

RTOS	Artefact	SHA-256 (truncated)
ChibiOS	ELF	`536da90f60dec5f9ed8c426ce39e31eddcc8bb41c08fd8fe...`
	MAP	`a1bf6840c22f1b189cd90b97484cfa8f99ba9ee71ddd122e...`
FreeRTOS	ELF	`a35ed9733625b2682749ee8ca7261db9e0a68e9fab7c392c...`
	MAP	`48ad955a91609eae007cf94381f66ca72693f91341145fed...`
Zephyr	ELF	`d60d8a90927c70c3f70cce025304f645fc708c24dc832c68...`
	MAP	`370fb4de8ecea42de68eaacb8dd482613f0de61fdde18324...`

Per-run validation status

RTOS	Profile	Run	Validated	clock (Hz)	VOS	FLASH_ACR	tickless	WFI
ChibiOS	fair_perf	01	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	02	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	03	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	04	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	fair_perf	05	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	01	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	02	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	03	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	04	yes	480000000	VOS0	0x00000034	OFF	OFF
FreeRTOS	fair_perf	05	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	01	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	02	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	03	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	04	yes	480000000	VOS0	0x00000034	OFF	OFF
Zephyr	fair_perf	05	yes	480000000	VOS0	0x00000034	OFF	OFF
ChibiOS	realistic_tickless	01	yes	480000000	VOS0	0x00000034	ON	ON

ChibiOS	realistic_tickless	02	yes	480000000	VOS0	0x00000034	ON	ON
ChibiOS	realistic_tickless	03	yes	480000000	VOS0	0x00000034	ON	ON
ChibiOS	realistic_tickless	04	yes	480000000	VOS0	0x00000034	ON	ON
ChibiOS	realistic_tickless	05	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	01	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	02	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	03	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	04	yes	480000000	VOS0	0x00000034	ON	ON
FreeRTOS	realistic_tickless	05	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	01	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	02	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	03	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	04	yes	480000000	VOS0	0x00000034	ON	ON
Zephyr	realistic_tickless	05	yes	480000000	VOS0	0x00000034	ON	ON

Conclusions

• **T1 - IRQ -> thread:** Under realistic_tickless (headline), ChibiOS showed the lowest median at 306 cycles, against FreeRTOS 846 and Zephyr 581. The fair_perf controlled baseline preserves the ranking (ChibiOS 297, FreeRTOS 440, Zephyr 624). The FreeRTOS rise from 440 to 846 (+406 cycles) is a measured cost of its tickless wake-up re-arm path under this published configuration; ChibiOS (+9) and Zephyr (-43) move much less.

• **T2 - Thread handoff:** ChibiOS showed the lowest median latency at 95 cycles, against FreeRTOS 376 and Zephyr 424. The measured median ratio in this test is approximately 4x, attributable to the lighter scheduler path of ChibiOS's native suspend/resume handoff to a higher-priority target.

• **T3 - Mutex uncontended:** ChibiOS 62 cycles is the shortest fast path; Zephyr 125 is second. The test measures mutex lock/unlock, so the FreeRTOS mutex API (xSemaphoreTake/Give on a statically-created mutex) is the correct primitive under test; its higher 195-cycle path reflects that FreeRTOS implements mutexes on its queue/semaphore core, which is intrinsic to the kernel and not a configuration the benchmark imposed.

• **T4 - Mutex contended + PI:** ChibiOS showed the lowest contended-path latency at 200 cycles, with FreeRTOS at 612 and Zephyr at 677. Priority inheritance is correct in all three kernels: 500/500 PI scenarios pass per RTOS per profile.

• **Reproducibility:** run_spread is zero cycles on every (RTOS, test) cell across the campaign. All 30 published run01..run05 captures (3 RTOS x 2 profiles x 5 firmware loads) passed the publication gate; the six run00 warmup captures are discarded.

• **Honest scope:** Phase 1 publishes the DWT cycle delta A4 - A1 for T1, not the hardware-event-to-thread external latency. The hardware-routed IRQ entry path (NVIC, stacking, vector fetch) is intentionally excluded and will be measured by the logic-analyzer in Phase 2 per ADR-015.

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This document is not legal advice; a final legal review is recommended before official publication.

FreeRTOS and Zephyr are excellent RTOS projects with different design goals. This comparison focuses only on real-time latency under the tested conditions.

Appendix A - Build commands

Each port has a single, deterministic build entry point. All ports honour the *PROFILE* selector (fair_perf, realistic_tickless, debug_dev) so that the same source tree produces the publication firmware variants without ambiguity.

ChibiOS

```
cd chibios/benchmark_chibios
make PROFILE=fair_perf -j
make PROFILE=realistic_tickless -j
```

FreeRTOS

```
cd freertos/benchmark_freertos
cmake -B build/fair_perf -G "MinGW Makefiles" -DPROFILE=fair_perf
cmake --build build/fair_perf
cmake -B build/realistic_tickless -G "MinGW Makefiles" -DPROFILE=realistic_tickless
cmake --build build/realistic_tickless
```

Zephyr (board stm32h750b_dk)

```
cd zephyr
.venv/Scripts/activate.bat (Windows)
. .venv/bin/activate (Linux)
west build -d build/fair_perf -b stm32h750b_dk benchmark_zephyr -p always -- -DPROFILE=fair_perf
west build -d build/realistic_tickless -b stm32h750b_dk benchmark_zephyr -p always --
-DPROFILE=realistic_tickless
```

Appendix B - GPIO mapping (logic-analyzer)

Six GPIO signals are exposed on the STMod+ connector P1 of the STM32H750B-DK for logic-analyzer capture. In Phase 1 they are routed by the firmware for parity with the future Phase 2 dual-source measurement and are not part of the published headline. See ADR-007 for the rationale behind the pin assignment.

Signal	MCU pin	STMod+ P1 pin	Role
A0_HW	PA0	1	TIM2 CH1 PWM mode 2 - hardware IRQ stimulus for T1
A1	PH1	17	LOW_LOCK - mutex lock by LOW (T4)
A2	PH4	19	HIGH_WAIT - HIGH blocked on mutex (T4)
A3	PH8	20	LOW_UNLOCK - LOW releases mutex (T4)
A4	PH12	11	HIGH_ACQUIRE - HIGH acquires mutex (T4)
MEDIUM_RUN	PI11	18	MEDIUM scheduled (PI excludes MEDIUM, T4)

Appendix C - Zephyr resolved configuration

Latency-relevant Kconfig symbols as RESOLVED in the generated *.config* for each publishable profile (not the raw *prj.conf* fragments). *not set* means the symbol is absent from the generated configuration. Snapshotted from *zephyr/build/<profile>/zephyr/.config* by *scripts/zephyr_config_snapshot.py*.

Kconfig symbol	fair_perf	realistic_tickless
CONFIG_MULTITHREADING	y	y
CONFIG_SCHED_SIMPLE	y	y
CONFIG_SCHED_SCALABLE	not set	not set
CONFIG_SCHED_MULTIQ	not set	not set
CONFIG_PREEMPT_ENABLED	y	y
CONFIG_NUM_PREEMPT_PRIORITIES	8	8
CONFIG_TIMESLICING	not set	not set
CONFIG_SYS_CLOCK_TICKS_PER_SEC	1000	1000
CONFIG_TICKLESS_KERNEL	not set	y
CONFIG_ARM_ON_ENTER_CPU_IDLE_HOOK	y	not set
CONFIG_PM	not set	not set
CONFIG_ASSERT	not set	not set
CONFIG_LOG	not set	not set
CONFIG_DEBUG	not set	not set
CONFIG_TRACING	not set	not set
CONFIG_THREAD_RUNTIME_STATS	not set	not set
CONFIG_THREAD_MONITOR	not set	not set
CONFIG_USERSPACE	not set	not set
CONFIG_HEAP_MEM_POOL_SIZE	0	0
CONFIG_SYSTEM_WORKQUEUE_STACK_SIZE	512	512